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Variability of Sea Ice During the 2007-2008 Canadian International Polar Year Program

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Abstract :

This article describes sea ice change and variability during the Canadian International Polar Year (IPY) program and considers some regional and hemispheric causes of this change. A companion paper explores the implications of this observed change and variability for ecosystem function, climate-related gas exchange, primary and apex predator habitat, and boreal populations. This shows an overview of the impact. The sea ice-themed research project as part of the 4th IPY is sure to be one of the most diverse international scientific programs. These have greatly improved the exchange of Inuit knowledge and scientific ideas between nations and disciplines. This interdisciplinary and cultural exchange will discuss the impact of the seasonal ice-free transition of the Arctic Ocean and ecosystem, the corresponding replacement of perennial to annual sea ice species, and the impact of this globally important sea ice species. It helped explain and convey the causes and consequences of perversion. can be broken by long-term waves generated in the vast open waters of the Chukchi Sea.

Keywords : International Polar Year (IPY), Hemispheric, Sea ice

Introduction :

The Anthropocene is an era of extraordinary change in the Arctic. We are experiencing unprecedented fluctuations in both the rate and magnitude of changes in the cryosphere, atmosphere, and lithosphere, which also depend on ecosystem functioning, as well as industrial development and the associated globalization of regional economies. is progressing. These changes are testing Canada's ability to respond as a polar nation and develop a coordinated, science-based policy toward the Arctic. It is becoming increasingly clear that general circulation model (GCM) predictions made 30 years ago are now becoming reality.

Satellite data shows that September sea ice extent has declined by about 10 percent every decade since the late 1970s, with a significant 65 percent decline in 2007 alone and a similar decline in 2011. Increased ring formation in the Northern Hemisphere accelerates storms, coastal erosion, and the input of warm Atlantic water into the Arctic basin. On land, a significant increase in temperature around years will have widespread effects on glaciers and permafrost, lake ice sheets and ice shelves (IPCC 2007). River flows in the Siberian Arctic are increasing due to the acceleration of the hydrological cycle.

The accelerating melting and seaward migration of the Greenland Ice Sheet increases the input of freshwater into the ocean, complicating freshwater-marine coupling in the surrounding ocean. Permafrost warming and degradation has been observed throughout the Arctic. A large amount of methane is released into the atmosphere, which will definitely improve the atmosphere in the greenhouse [1].

These recent changes strongly support the idea that the Arctic is warming much faster than even the most pessimistic numerical simulations predict (IPCC 2007). All indications are that the Arctic climate is heading towards a new equilibrium.

The rate of this transition is unlike anything seen before in the Anthropocene and is likely the result of anthropogenic impacts on the atmosphere. In this paper and its companion paper, we examine the concept of "ice landscape." To do this, we make use of the similar term "landscape", which is well defined in the scientific literature.

Materials and methods:

Traditional knowledge of sea ice

The IPY-CFL project was based on the concept of two ways of knowing. This is a concept that recognizes that there are two equally valid approaches to gaining knowledge of the Arctic environment that helps us understand the effects of climate change on fault guidance systems. Previous research in Inuvialuit settlements that Inuvialuit people are already seeing and experiencing the effects of climate change.

The region's long history of occupation provides opportunities for residents to study visible changes in sea ice topography, weather patterns, and animal movements, and the impact of those changes on culture and the environment [2].

A key element of the IPY-CFL project was traditional knowledge research aimed at documenting Inuvialuit cultural knowledge and sea ice use (cultural icescapes), as well as observations related to climate change. IPY-CFL project members (CFL Team 10) interviewed a total of 49 Inuvialuit Document knowledge about sea ice and the factors that influence sea ice change.

These 49 community members have a combined total of approximately 2,290 years of track and field experience, averaging approximately 47.7 years of experience per person. The research team also investigated how changes in sea ice affect marine ecosystems and resources, and how Inuvialuit adapts to such changes (figure 1).

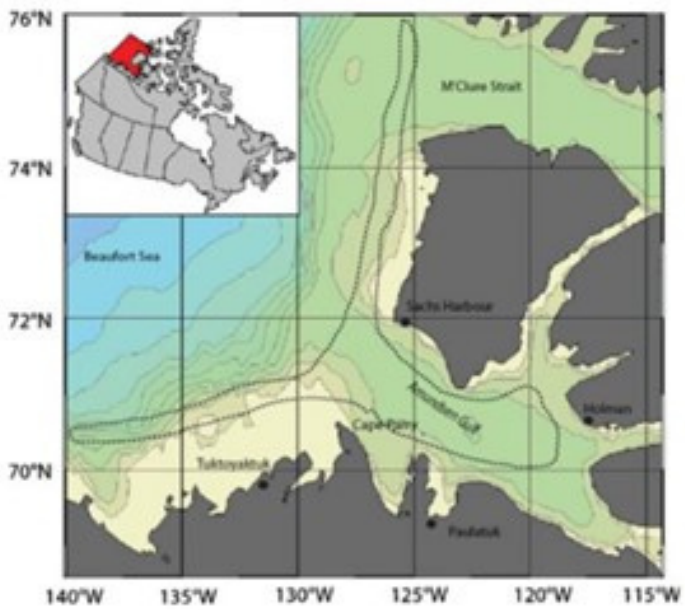

Figure 1: The Circumpolar Flaw Lead System Study area and Inuvialuit communities.

Forty-nine respondents from three communities spoke about the importance of sea ice, including the important properties of lead from the Cape Bathurst Fault and the important role lead plays in wildlife and subsistence hunting in Inuvialuit. We talked. Open bodies of water and ice cracks (pipes) are also important sites for seals, polar bears, and humans. Seals use ridges to build dens and swim in open water to feed.

The polar bear approaches the seal by following a crack in the ice. Inuit hunters use pressure combs to hunt and set nets. Further details are provided in the accompanying paper on the effects of sea ice change during IPY [3].

The most common observations reported by respondents across communities were decreased sea ice and increased open ocean. During the winter, when previously it remained frozen, bodies of water became wider and they appeared more often. Open bodies of water were also closer to shore than usual, suggesting changes in rapid ice formation. Respondents from Paulatuk Township said they typically see open water north of Cape Parry and Pierce Point in October and November, but that in the waters near Paulatuk, there are no cracks along the coast that open after freezing, said there is usually no open water in the bay during these months. Sachs Harbor community members also note that freezes are occurring about a month later than they were 20 to 30 years ago [4].

Respondents noted that regional ocean currents and wind strength determine the state of the open ocean and the size and frequency of sea ice divides and cracks. For example, near Cape Parry, ocean currents influence the formation of ice diversions and cracks. Sea ice conditions are primarily determined by the interaction of winds with existing open ocean and ice cracks. Strong winds have been identified as a major factor increasing the formation of sea ice cracks and slips. This also typically occurs during times when the ice is less likely to break (e.g., January, February, etc.). Wind direction has also been confirmed to be an important factor in sea ice runoff.

More open water and ice cracks can have serious impacts on travel and hunting, and can have many cultural, economic, and physical impacts. Respondents discussed how difficult it is to hunt in open waters. Respondents in Sachs Harbor said traveling over sea ice deep into Amundsen Bay was no longer considered safe. Increased exposure of open water and sea ice also affects wildlife. For example, polar bears are harder to spot. Additionally, the expansion of open ocean areas has made it difficult for polar bears to approach seals [5].

Western scientific knowledge about sea ice:

The Western scientific approach to two types of knowledge has also generated many new insights into the causes of sea ice change and variability. Sea ice acts as an insulating layer between the atmosphere and ocean by providing a barrier to heat and mass transport across the ocean-sea ice-atmosphere interface (OSA). The albedo (reflectivity) of sea ice terrain makes it an important component of the Earth-ocean-atmosphere system. Sea ice reflects up to 50% of the incoming solar radiation (albedo is approximately 0.5). On the other hand, snow-covered sea ice reflects up to 90% of the incoming solar radiation (with an albedo of approximately 0.9) [1].

The high reflectivity of sea ice maintains both cold Arctic temperatures during the solar season and the equatorial and polar temperature gradients that govern thermal circulation at the hemispheric scale. The dramatic decline in Arctic sea ice is having a significant impact on the Earth's energy balance. The latter is defined by the balance between incoming and outgoing solar radiation.

Sea ice extent is defined as an area with an ice concentration of at least 15% based on passive microwave satellite records. In the Northern Hemisphere, the largest historical extent occurs in March, when sea ice covers approximately 15 million km2. Sea ice extent is at its lowest in September, when sea ice covers approximately 7 million km2.

In recent years, the minimum extent of sea ice in summer has decreased significantly. highlights the minimum sea ice extent observed in 1979, 2000, 2005, and 2007, and compares the average sea ice extent from 1979 to 2000 with the observed minimum sea ice extent in 2000, 2005, and 2007. We are comparing the minimum sea ice extent (figure 2) [6].

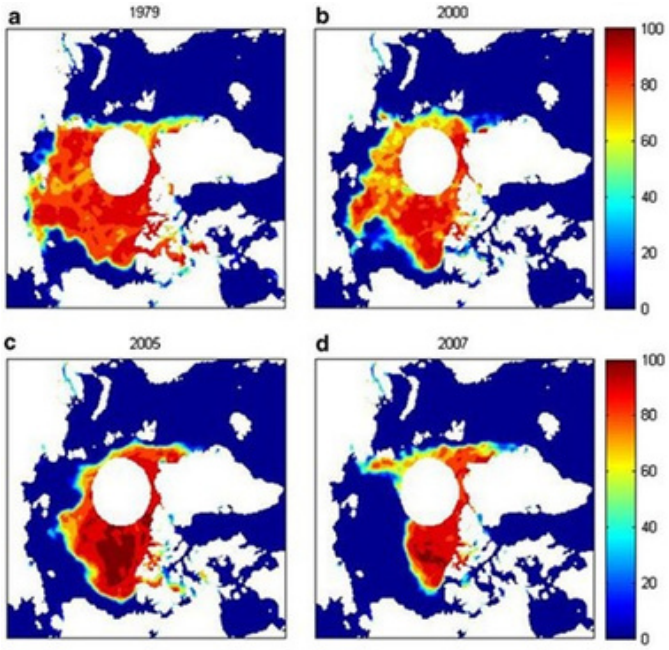

Figure 2: Sea ice minimum extents (% sea ice concentration) (a) 1979 (start of the satellite record), (b) 2000, (c)2005, and (d) 2007 (record minimum summer sea ice extent). Note: data missing north of 87.5°N due to limitations of polar orbiting satellite paths.

Study of Summer Sea Ice Loss This magnitude results in negative trends in annual and seasonal cycles with annual averages showing a statistically significant overall trend of decades. -42,100 km2 per area. Including the 2008-2010 melt season, the decline rate increases to -81,310 km2 per decade (-12.4% per decade), and 2011 is about the same as 2007 so this proportion may increase further if the summer 2011 minimum range is included in the trend (NSIDC 2011).

The most dramatic declines in regional sea ice extent have occurred in the Kara and Barents Seas, and more recently in the Beaufort and Chukchi Seas [4]. There is. Acceleration of sea ice loss in the 21st century has resulted in a significant decrease in ice thickness. Previous studies have found that the most significant declines occur in ice marginal zones and coastal regions. Other studies have also shown regional differences in ice thickness and significant ice loss over multiple years (figure 3) [7].

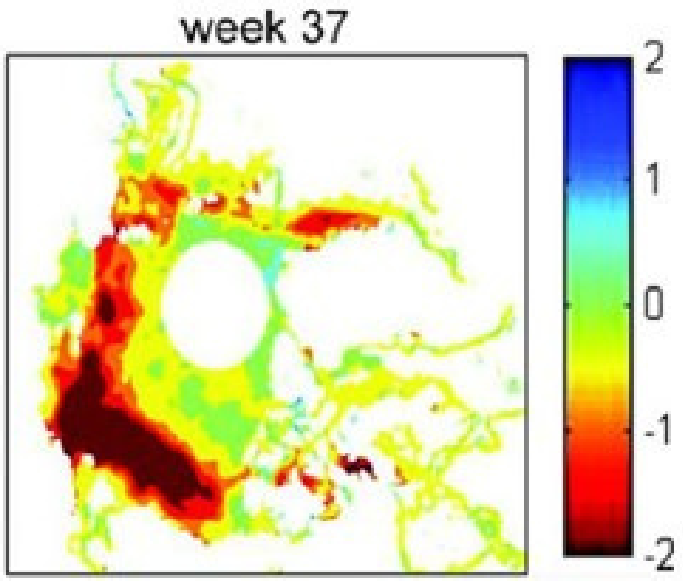

Figure 3: Spatial-temporal trends in sea ice decline for week 37.

Sea ice status in the Canadian Arctic Islands:

Recent studies have revealed variability in ice concentrations in the Canadian Arctic Islands (CAA). Statistically speaking, reported a significant decrease in mean total sea ice area in September of 8.7 µcade-1. This is in contrast to the statistically insignificant trend. They reported changes in sea ice extent in the CAA on the order of -5000 ± 400 km2 per decade, with the strongest trend observed in October. September MYI area within CAA decreased at a rate of 6.4 µcade-1. However, this trend remains statistically insignificant due to increasing dynamic MYI imports from the Arctic Ocean.

The observed decrease in sea ice area in the CAA can be attributed to a longer melt season and sea ice dynamic processes (sea ice flux). The duration of the thaw phase within the CAA increases significantly by approximately 7 days per year. The longest thaw period on record was reported in 2008, with a duration of 129 days. This is 13 days longer than the average duration of the melting season of 116 days. A study of sea ice flow from 2002 to 2005 found that the largest ice exports and imports occurred in the McClure Strait, Amundsen Bay, and Lancaster Bay regions, with an area of 122 × 103 km2 between September 2002 and June 2007. showed a net loss of sea ice. MYI ice may continue to exist in McClure Strait and the western region of the Northwest Passage during the transition to an ice-free summer Arctic [8].

Compares the extreme sea ice minimum observed in his CAA in 2007 with his 1998 one. The mean September sea ice extent was 2.90 and 2.65 standard deviations below the 1968–1996 historical climate in 1998 and 2007, respectively, and the October sea ice extent in 1998 and 2007 was below the 1968-1996 historical climate, respectively. It was 4.45 and 3.36 standard deviations below the level.

Sea ice status in Baffin Bay:

Negative sea ice concentration anomalies exist in southern Baffin Bay in winter and spring and throughout the basin in summer, indicating a tendency for seasonal sea ice area to decrease. I am. Meteorological conditions and ocean currents determine sea ice in Baffin Bay, with ice velocities being greatest in the north. The North Water (NOW) Polinya in northern Baffin Bay transfers heat and radiation between the atmosphere and ocean. The result is an area of significant biological productivity. The North Water Polinya expands in March and reaches its maximum area in July. In recent years, there has been a negative trend in sea ice concentration in the northern region of Baffin Bay, suggesting that the polynya is expanding faster. This is complicated by the formation of an ice bridge north of NOW Polinya, which is interfering with the Lincoln Sea ice. If ice bridges do not form, modern polynya would be dominated by ice export to this region, making it a marginal ice zone rather than the usual open water of polynya. Weekly trends show the greatest decrease in sea ice extent in early November [9].

Hudson Bay sea ice status:

Investigated trends in fall sea ice concentration and sea ice extent in Hudson Bay over the period 1980-2005. Using data from the Canadian Glacier Service, sea ice concentration anomalies exhibit statistically significant trends ranging from -23.3% to -26.9% per decade. The average sea ice area based on seasonal ice concentration (SIC) of over 80% decreased by 1.05×105 km2 for every 1°C increase in temperature at the end of November, to -1.17×105 km2.

The average decrease in sea ice extent during the satellite record (1980-1995 vs. 1996-2005) ranged from 112,000 km² to 241,000 km² depending on the week of the year. Changes in regional surface air temperature during freezing periods are associated with observed changes in hemispheric teleconnections also investigated spring temperature and sea ice trends.

They showed that spring surface temperature anomalies around Hudson Bay increased by 0.26 to 0.30°C per decade between 1960 and 2005. Negative trends in sea ice extent correspond to positive temperature anomalies. Sea ice concentration trends ranging from -15.1 to -20.4% per decade were observed in western and southwestern Hudson Bay.

Hudson Bay sea ice area decreased by 8,790 km2 per year to 10,035 km². It was found that the extent of sea ice in spring strongly depends on the surface air temperature around Hudson Bay in the spring and the previous fall, and on the sea ice dynamic forcing expressed by the relative vorticity of the Hudson Bay ice [10].

Sea Ice Status in the Beaufort Sea:

The influence of a receding ice sheet on the climate of the Beaufort Sea region was investigated in a recent assessment of the region's meteorology, which found that autumn temperatures have increased by more than 6°C. Therefore, sea ice in the Beaufort Sea region in August 2009 consisted mainly of broken ice or floating ice in advanced stages of breakup.

This resulted in a significant decrease of sea ice in summer 2010 (NSIDC 2010) and also in 2011. In fact, ice maps of the western Arctic showed a decline in the distribution of thick multiyear ice in the Beaufort Sea in September 2010. Of particular note is the rarity of thick MYIs, which existed only as thin bands (red areas) interspersed with areas of relatively low first-year ice concentration.

The evolution of sea ice cover in Amundsen Bay and the southern Beaufort Sea was assessed in the IPY-CFL study (September 1, 2007 to September 1, 2008). Sea ice extent was assessed by ice type using weekly digital ice map data for the western Arctic obtained from the Canadian Glacier Service digital archive.

The data are gridded (1 pixel 02 × 2 km) and sea ice concentrations are extracted directly from the CIS data for old sea ice, thin first-year sea ice, new sea ice, and total sea ice (100 km of sea ice cover). displayed in fractions of a minute). Ice maps are created by expert interpretation of satellite images following the CIS classification protocol. For each digital ice map, a superset of older 1-year ice maps: thick 1-year ice map (>120 cm), medium 1-year ice map (70-120 cm), and thin 1-year ice map (30 cm) percentage of total -70 cm)), young ice (10-30 cm) new ice fraction [11].

The largest decrease in sea ice concentration in the Beaufort Sea during the 2007-2008 IPY occurred in the southern Beaufort Sea (Canadian waters and northern Alaska) from week 33 to week 41 (July-August). These observations are consistent with previous studies showing that the greatest decreases in sea ice extent in the Beaufort Sea occurred in summer and fall. In particular, losses on the order of -10,100 ± 2,200 km per year have been calculated for the region including the Beaufort Sea and the Arctic Ocean.

Climate forcing of sea ice:

Climate change is the result of both natural and anthropogenic forcing. Global warming has accelerated in recent decades due to increased air pollution from greenhouse gases. Some of the strongest signals are seen in the Arctic, particularly with the dramatic decline in sea ice since 1979. The Intergovernmental Panel on Climate Change (IPCC 2007) report highlights the fact that certain elements of the Earth system, such as sea ice, undergo rapid transitions as the transition from liquid to solid phase is temperature dependent. Did. Arctic surface temperatures have increased by 1-3°C over the past decade, and Arctic sea ice extent and thickness have declined at an accelerating rate, highlighting the dramatic effects of warming in high latitudes. Masu. The unprecedented accelerated sea ice change in the early 21st century has been termed the “recent Arctic warm period”. Previous studies have shown that rising surface atmospheric temperatures are responsible for the increase in ocean heat content and latent heat in the Arctic Ocean, which is associated with a decrease in sea ice extent. Satellite measurements now show that warming has accelerated in recent decades compared to 100-year trends, with autumn warming being greatest in the Chukchi and Beaufort Seas. An average extension of the thawing period by 10 to 12 days was also observed [11].

This trend has recently been extended to show that the current sea ice decline is unprecedented in at least 1,450 years, based on paleoclimate reconstructions of Northern Hemisphere sea ice extent. Changes in sea ice are associated with an enhanced ice albedo feedback mechanism, reducing the albedo of the Arctic surface cover. The mechanisms responsible include (1) increased surface temperatures in recent decades, (2) large-scale atmospheric forcing mechanisms and meridional transport that affect increasingly mobile ice masses in the Arctic, and (3) fluctuations in cloud cover and the associated decline in radiation, (4) increased solar absorption due to increased ice cover in the first year.

Recent studies show that unprecedented sea ice melt in the Beaufort Sea in 2007 was due to increased ocean heating due to increased open ocean, solar heat input, and poleward advection of ocean heat flux through the Bering Strait, thereby strengthening the ice albedo feedback mechanism responsible for accelerated ice loss. This phenomenon may also contribute to an increase in open water and ice that is severely rotten in the first year, termed “rotten ice” [12].

In addition to sea ice growth and decay, the dynamic properties of sea ice, i.e. its movement, play an important role in the Arctic ice landscape. Open leads occur especially when the ice mass diverges and cracks appear in the ice. In contrast, when ice floes converge, a pressure bulge occurs. Spatially heterogeneous ice loss in the Northern Hemisphere is thought to be due to dynamic fluctuations in ice export from the Western Arctic to the Eastern Arctic. Summer ice advection from the Pacific Ocean to the Atlantic Ocean was found to be responsible for the decline in summer sea ice extent from 2003 to 2007, decreasing from 7% to 21% in 2003 and 2006, respectively.

However, in 2007, advection increased, found that during the summer of 2007, a persistent sea-level high over the Canadian Basin accelerated ice export from the Pacific to the Atlantic, reducing cloud cover and increasing down welling radiation [3]. The third minimum sea ice extent was observed in September 2009. Increased surface temperatures in the Canadian Arctic Islands in spring 2010 also contributed to significant ice loss in the region, but in summer 2010 a weakening of the atmospheric dipole pattern in sea level pressure re-emerged and This resulted in a poleward retreat of sea ice. Sea area (NSIDC 2010). As permanent ice disappears, sea ice cover becomes increasingly fragile and more vulnerable to contributions from changing thermodynamic and dynamic regional atmospheric forcing mechanisms.

Ice eddy circulation in the Beaufort Sea is traditionally characterized by anticyclone circulation, which reverses to cyclonic circulation in summer. However, recent studies have shown that the Beaufort Sea ice eddy reverses during its annual cycle, and such reversals have occurred more frequently in recent years.

Investigation of ice eddies in the Beaufort Sea using 22 ice beacons launched between November 2007 and May 2008 reveals small features in sea ice movement in the southern Beaufort Sea. The results of this analysis demonstrate different scaling laws (power-law We have demonstrated that the following rule exists. Winter Consistent features of sea ice movement were also expressed by loops and meanders determined by local wind-driven phenomena in winter and non-local phenomena in spring [2,6].

The ice mass that decreases in summer becomes more mobile and vulnerable to periodic Beaufort vortex inversions caused by cyclones. This can promote ice divergence and lead to increased melting rates. studied the synoptic climatology of atmospheric forcing and its relationship to sea ice movement and Beaufort eddy inversion. His catalog of 12 daily synoptic weather types for the Beaufort Sea region for the period 1979-2008 using NCEP/NCAR reanalyses (meaning sea level pressure data) has been created. Synoptic types are constructed using a mixed method using principal component analysis and K-means clustering to describe the surface pressure climatology of the southern Beaufort Sea region over the period 1979-2008. Synoptic Types 1, 6, and 7 are associated with arctic cyclones, and all other types are associated with anticyclones. Well-defined Beaufort highs are represented by types 2 and 5. Furthermore, types 1, 3, 4, and 8 were shown to occur in summer [13].

We found a significant lag correlation between synoptic type and 12-week smoothed sea ice eddy strength, indicating that the type of cyclones that promote southerly or eastward atmospheric circulation in the southern Beaufort Sea Typically It has been shown to precede reversal by 2 to 10 weeks. If ice extent continues to decrease, Beaufort Circulation inversions due to Cyclone may occur more frequently throughout the year, leading to increased winter lead formation and ice formation during the annual cycle. may lead to divergence.

This may represent a positive feedback during the summer, as increased ice divergence reduces sea ice concentration and areal albedo, increasing absorption of descending solar radiation and thereby increasing melting. Conversely, negative feedbacks can occur during winter when new ice forming on icebergs is easily compressed by existing ice floes into thick ice ridges, thereby increasing ice thickness [9, 13].

Discussion:

In a follow-up study, investigated the relationship between sea ice cover in Amundsen Bay and the overlying atmospheric boundary layer over weekly time steps from winter to summer 2008. Total sea ice cover (approximately 97% of sea ice) from January 7 to April 21 (ice sheets with 3% lead content). From April 28th to May 12th, a total of sea ice areas approached 100%. After May 19, total sea ice area rapidly decreased to about 3% of its July low. In winter, a thermal internal boundary layer was present in 10 of the 12 daily mean temperature profiles.

The average depth was 375 m below the radiation inversion reaching approximately 1100 m. This profile is consistent with the typical vertical thermal structure on Arctic sea ice described by Overland and Guest (1991). Thermal IBL is the mesoscale upward flux of sensible heat (range 17.0-70.7 Wm-2, mean 0.45.7 Wm-2)) - ice sheet guidelines for open water or new sea ice, and young sea ice [thin ice ≤ 0.3 m]). Its area-averaged surface temperature was generally higher than the temperature near the surface of the approaching air mass. Thermal IBL was also present in eight of the nine mean diurnal profiles in early spring. However, surface fluxes have not been estimated for this period. From late spring to early summer, stable IBLs existed in seven of the eight temperature profiles. The average depth was 250 m below the remaining mixed layer extending to approximately 700 m. The stable IBL results in a mesoscale downward flux of sensible heat (range -1.6 to -45.1 Wm-2, average 0-18.8 Wm-2) was considered to be the cause. %). (July), regional average surface temperatures are generally lower than temperatures near the surface of the approaching air mass. The presence of IBL, which was common during the winter and summer of 2008, is due to the fact that the surface layer of the atmosphere reacts to unconsolidated ice floes by a persistent lead network that exists between ice sheets during the winter and early spring. This suggests that it was a reaction to the sea surface. From late spring to summer after demolition. However, during the Arctic night, increased inversion over the IBL buffered the surface layer and sea ice surface from the free atmosphere.

Conclusion:

The findings of Canada's IPY program illustrate the breadth and depth of the linkages between sea ice landscapes and the processes that influence this ice landscape. The results of this article can be used to assess the sea ice situation in the Arctic basin and predict the future. Policy makers need to consider many interrelated physical processes and characteristics of sea ice, such as sea ice divergence/convergence, changes in sea ice movement, and seasonal ice extent reduction; Not limited to these. This will affect local populations and the use of sea ice for hunting and transportation activities. These impacts extend across the Arctic, affecting commercial shipping, shipping, and aspects of the ecosystem such as polar bear hunting, nutrient transport, and the 4,444 pollutant pathways within this system.

Significant reductions in sea ice extent and thickness are already having significant impacts on northern coastal communities. Increased storm surges due to a decrease in the extent and duration of sea ice in coastal regions are increasing coastal erosion, accelerating the movement of human infrastructure, and impacting fishing and other harvesting activities.

Seasonally changing sea ice extent and the timing of freezing and breakup affect the safety and efficiency of transportation for residents and hunters in northern regions, as well as the food security of populations throughout the annual cycle. To ensure the adaptive capacity of northern communities, the demands posed by the physical drivers of climate change need to be addressed in the context of socio-economic factors and globalization (ACIA 2005; IRPP 2009).

Ice fluctuations and loss, and changes in sea ice flow between the Arctic Ocean, Baffin Bay, and the Canadian Arctic Islands, have important implications for Arctic transport. Once largely inaccessible to commercial shipping, the Northwest Passage has been seasonally ice-free since 2007. Warmer spring temperatures and increased multiyear ice loss (NSIDC 2010) led to record sea ice loss in the Northwest Passage in August 2010. This has important and significant implications for seasonal transportation, industrial development, navigation, and sovereignty of the Canadian Arctic.

Conflict of interest:

Author declares there is no conflict of interest.

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